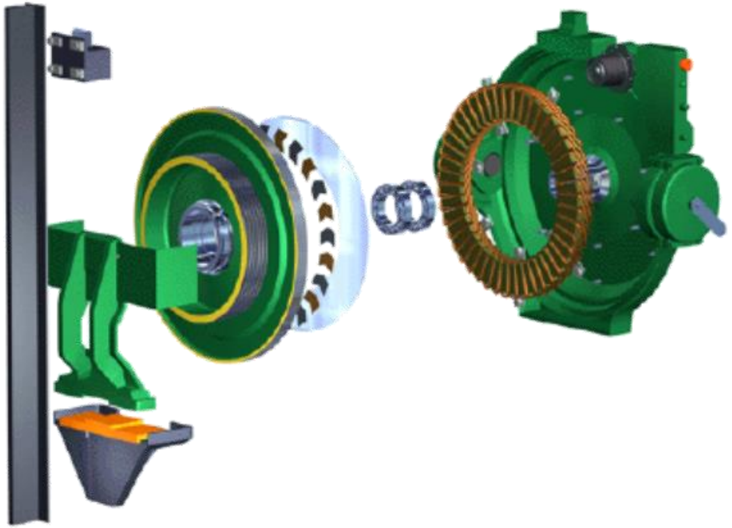
Monarch solution for KONE EcoDisc machine in retrofit project

Ian

12th,July,2017

KONE EcoDisc machine introduction

KONE EcoDisc motor was first introduced in KONE MonoSpace elevator, world's first MRL elevators (in 1996)



Since then, it has been expanded to most of KONE elevators benefiting from its energy efficiency, less weight, easy installation and good ride comfort.

Over 400,000 EcoDisc machines have been installed worldwide as of 2012– data released from KONE
(As of today, it is estimated that over 800,000 have been installed, quarter of them in China)

KONE EcoDisc machine by size

MX 05 This is the smallest EcoDisc machine

MX 06 A bigger EcoDisc machine compared to MX 05

MX 10 the most common EcoDisc machine and it is used in the MonoSpace elevators.

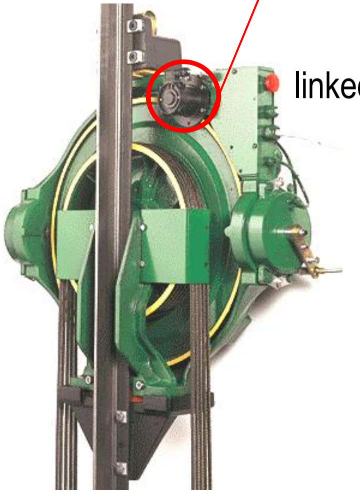
MX 18 mainly used in MiniSpace and possibly EcoSystem MR as well as some modernized elevators by Kone.

MX 20 A large and more powerful EcoDisc machine compared to MX 10. It is mostly used in the MiniSpace MR (2:1 roping) or TranSys MRL (4:1 roping) elevators.

Why it is a challenge when driving EcoDisc machine

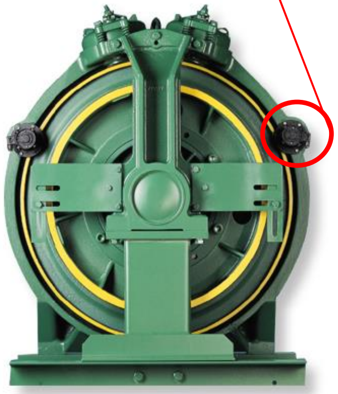
Tachometer

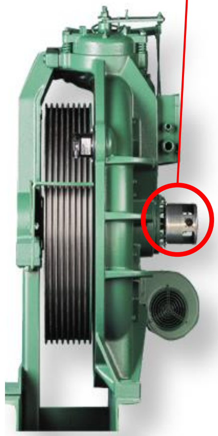
linked to friction gear.



Model like MX05/06/07/10/11/15

Tachometer + Transformer





Sinusoidal voltage

Model like MX18

Most current drives can not support KONE EcoDisc machine with Tachometer or Transformer directly

Monarch solution for EcoDisc machine

Encoder: Monarch tailored incremental encoder to replace the original tachometer

before

after

Encoder main features

1.Easy installation

Same mounting dimension as original

2.5Vdc power supply, 1024 PPR, ABZ

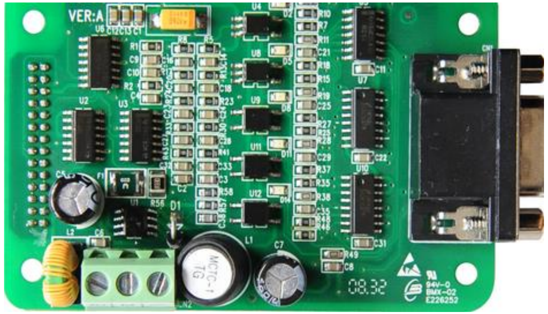
More than 20,000 pulse per motor rotary

3.Encoder package model

MBT-EC-MX

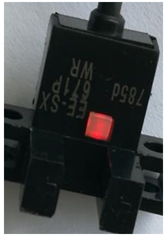
Monarch solution for EcoDisc machine

New design driving control algorithm and PG card



PG card model: MCTC-PG-D1

Tailored algorithm to control EcoDisc with incremental ABZ encoder.



Parameter setting and motor tuning

Motor tuning performed following the normal PMSM operation process for PMSM, except the parameter setting difference comparing with the PMSM motor tuning.

Parameter code	Parameter name	Setting description
F1-00	Encoder type	2: ABZ encoder
F1-01~F1-05	Motor data	Set as per actual motor data
F1-11	Tuning	1: motor tuning with load 3: lift shaft tuning
F1-12	Encoder PPR	default as 20,000, update automatically after motor tuning
F1-25	Motor type	1: PMSM

Note:
During motor tuning, motor will rotate forward then downward. Release the inspection up/down button after it got initial rotator angle.

Faulty diagnostics
1.Err20, sub code 96
friction gear wear problem, checking friction gear or change new.
2.Err20, sub code 97
if coming during tuning, checking “Z” signal from the photoelectric sensor; or friction gear problem.

Project view

Machine room type

Speed : 1.75m/s

Roping: 2:1

Load: 800kg

Floors: 18

Machine : EcoDisc MX10

Project view

Machine room less

Speed : 1.0m/s

Roping: 2:1

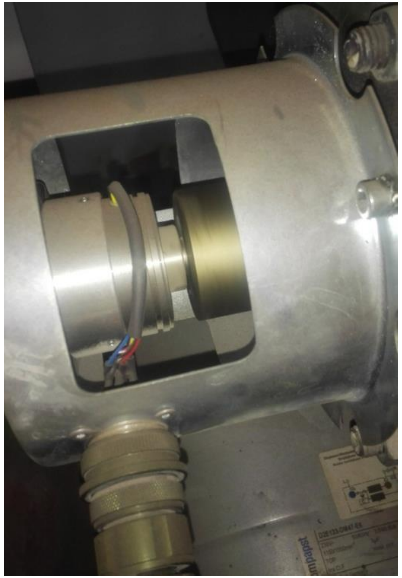
Load: 1000kg

Floors: 4

Machine : EcoDisc MX10

Alternative solution, only for MX18-project view

TAMAGAWA sin/cos encoder with Monarch tailored shaft coupling



before

after

Machine room type

Speed : 2.0m/s

Roping: 2:1

Load: 1350kg

Floors: 21

Machine : EcoDisc MX18

Forward, Always Progressing!